Statistical Thinking for Data Science

Week 6: Pictures that argue, and pictures that lie

Every graph is a comparison, so choose the comparison first

Dr Abdulzeid Yen Anafo

Department of Computing and Data Analytics
University of Mines and Technology, Tarkwa

First Semester, 2026/2027

Week 6: Pictures that argue, and pictures that lie

Learning Objectives

By the end of this week you should be able to:

- state the comparison a graph is making before you draw it, and judge whether the graph makes that comparison easy;
- rank the visual channels by how accurately people read them, and use the ranking to choose a chart;
- choose a display from the question and the type of the variables, rather than from habit;
- name the common ways a graph misleads, and say in which direction each one pushes the reader;
- show a distribution rather than a bar of means, and show uncertainty when it matters;
- design a colour scheme that survives greyscale printing and colour blindness;
- apply a short checklist before any graph of yours is seen by anybody else.

6.1 A graph is an argument

Last week we described relationships with numbers. This week we describe them with pictures, and the discipline is the same: a summary that hides what matters is worse than no summary at all, because it is believed.

Every graph answers a question of the form *compared to what?* Availability of 97%, compared to what? To last month, to the other plant, to the contract target, to the best shift. Until you can finish that sentence, you cannot choose a chart, and a chart chosen before the comparison is decoration.

The first question

Write the sentence you want the reader to be able to say after looking. If the sentence is "availability has drifted down for three months, from 97.4 to 96.2", the graph must put those four numbers side by side on one scale. If no sentence comes, you do not yet have a graph, you have a dataset.

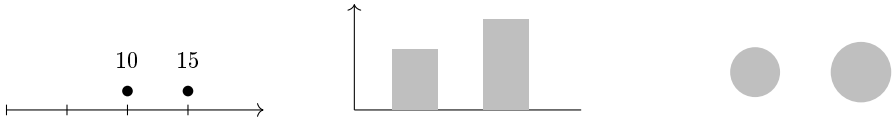
There are two different jobs, and confusing them wastes time. An **exploratory** graph is for you. It is quick, ugly, and you will draw forty of them. A **communication** graph is for somebody else, usually somebody who will look at it for eleven seconds and then decide something. Polish belongs to the second kind only, but honesty belongs to both: the fastest way to fool yourself is to draw a careless exploratory graph and believe it.

6.2 How people read a picture

A graph turns numbers into things the eye can measure: positions, lengths, angles, areas, shades, colours. These **visual channels** are not equally good. Decades of experiments, beginning with Cleveland and McGill in the 1980s, put them in a fairly stable order of accuracy:

1. position along a common scale (dots or points on one axis);
2. position along identical but separate scales (small multiples);
3. length (bars starting from a common baseline);
4. angle and slope;

5. area;
6. shading and colour saturation;
7. colour hue, which carries category, not quantity.



position on a common scale length from a baseline area

Each panel above encodes the same two numbers, 10 and 15. The ratio is obvious on the left, clear in the middle, and nearly invisible on the right. That is the whole case against pie charts, bubble charts and anything three dimensional: they spend the reader's attention on channels the eye reads badly.

The area trap

If you encode a quantity as the *radius* of a circle, doubling the quantity makes the circle four times as big, because area grows with the square of the radius. To double the apparent size you must multiply the radius by $\sqrt{2} = 1.41$. Most bubble charts drawn in a hurry are wrong in exactly this way, and they always exaggerate.

6.3 Choosing a display

Choose from the question and the variables, not from the chart menu.

What you have	What usually works
One numeric variable	histogram, or a dot plot when n is small; box plot only for comparing many groups
One categorical variable	bar chart with bars *ordered by value*, not alphabetically; never a pie chart with more than three slices
Two numeric variables	scatter plot, with a smoother if the pattern is not obvious
Numeric over time	line chart, with time on the horizontal axis
Numeric by group	dot plot or strip plot of the raw values, or box plots side by side; a bar of means only when the distribution is genuinely uninteresting
Two categorical variables	grouped bars of *conditional* percentages, or a mosaic plot
Many groups at once	small multiples: the same small chart repeated, same axes

6.3.1 Order the categories

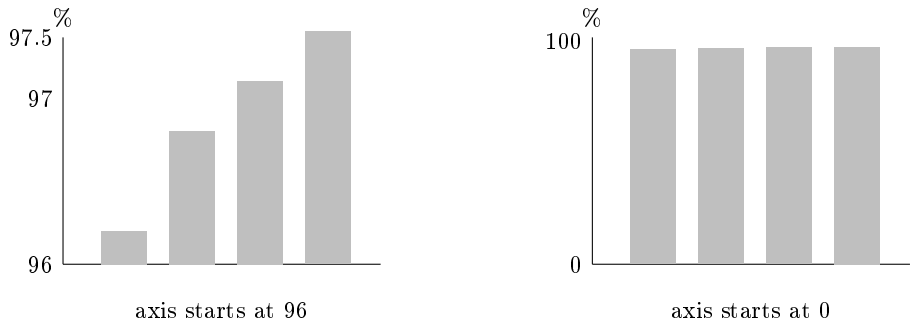
Bars in alphabetical order make the reader do the sorting. Order them by the value being shown, unless the categories have a natural order already, like Monday to Sunday or primary to tertiary. This single habit improves more graphs than any other.

6.3.2 Small multiples beat one crowded chart

Six lines of six colours on one axis is a puzzle. The same six series drawn as six small panels, each with the same axes and one line, with the overall pattern faintly in the background of each, is read at a glance. The cost is space; the benefit is that the comparison is actually made.

6.4 Six ways a graph misleads

6.4.1 The truncated axis



axis starts at 96 axis starts at 0

The same four months of plant availability: 96.2, 96.8, 97.1 and 97.4 per cent. On the left the last bar is four times the height of the first, and a board will read "dramatic improvement". On the right you see the truth, which is that availability moved by 1.2 percentage points.

Neither picture is a lie by itself. The left one is dishonest only if the caption does not say that the axis is cut, and the right one wastes the page if the question is about a one point change. The rule is: **bars must start at zero**, because the bar encodes length, and a cut baseline makes the length meaningless. A line or a dot plot may start anywhere, because it encodes position, as long as the scale is labelled clearly.

6.4.2 The bar of means

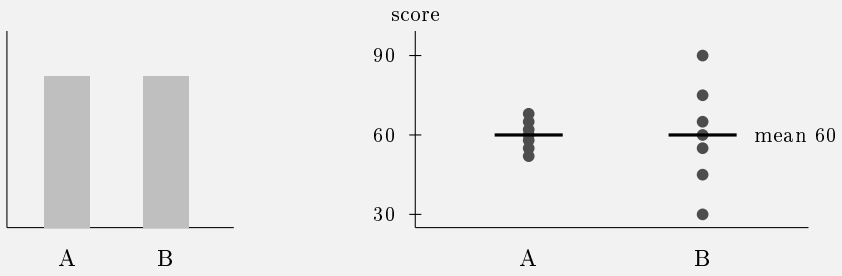
Example 1: two groups with the same mean

Two tutorial groups sit the same test, seven students each.

Group A	52, 55, 58, 60, 62, 65, 68
Group B	30, 45, 55, 60, 65, 75, 90

Solution

Both groups total 420, so both have a mean of exactly 60. A bar chart of the two means is two identical bars, and it says the groups are the same. They are not: the standard deviations are 5.6 and 19.6. Group A is uniformly middling; group B contains both the weakest and the strongest students in the room, and those two students need opposite things from you.



two bars of means the same data, plotted

The rule

With fewer than about fifty observations per group, plot the observations. The mean can go on top as a line. A bar chart of means throws away everything you were paid to find, and it is the single most common bad graph in student projects and in industry reports alike.

6.4.3 The cherry-picked window

A series that rises for three years and falls for the last six months can be made to tell either story by choosing where the axis starts. Ask always: why does the graph begin in that year? If the answer is "because that is when the data start", fine. If the answer is "because that is the lowest point", you are reading an advertisement.

6.4.4 Two vertical axes

A chart with a left axis and a right axis lets you slide one series up and down until the two curves appear to move together. The apparent relationship is manufactured by the choice of scales, and a different choice produces a different story. If you want to show two quantities in different units moving together, use two small panels, one above the other, with a shared time axis, or plot each as an index relative to its own starting value.

6.4.5 Area, volume and the 3D pie

Doubling a quantity and drawing a circle with twice the radius exaggerates by a factor of four; drawing a 3D cylinder exaggerates further and tilts it so the slices at the front look bigger. Perspective adds nothing a flat chart lacks. There is no situation in a technical report in which a three dimensional pie chart is the right answer.

6.4.6 No uncertainty anywhere

A bar chart of means from 8 observations per group, drawn with no indication of spread, invites the reader to believe a difference that a second sample would reverse. Show the data, or show an interval, and say what the interval is: a standard deviation, a standard error and a confidence interval are three different things with very different widths. We make this precise in Week 9.

Which way does each one push?

Truncated bars exaggerate change. Bubble radii exaggerate growth. Cherry-picked windows exaggerate whatever the author believes. Bars of means hide variation, which usually flatters the group with the worst students. Dual axes manufacture relationships. Missing uncertainty makes everything look certain. Every one of these errors is in the direction of a stronger claim, which is exactly why they survive.

6.5 Scales: when to use a log axis

If a quantity grows by a roughly constant *percentage*, a linear axis buries the early years and a logarithmic axis tells the truth. On a log scale, equal distances mean equal ratios, so constant growth appears as a straight line.

Example 2: a doubling series

A quantity reads 1.2, 2.4, 4.8, 9.6 and 19.2 in successive years.

Solution

Each year is exactly twice the one before. On a linear axis the first two years are nearly flat against the last one. Taking logarithms base 10 gives 0.079, 0.380, 0.681, 0.982, 1.283, and the differences are 0.301 every time, which is $\log_{10} 2$. On a log axis the five points lie on a perfectly straight line, and the slope is the growth rate.

Use a log scale when ratios are the meaningful comparison: epidemic curves, gold price, ore grade over orders of magnitude, income. Do not use one to hide a disagreeable pattern, always

label the axis so the reader knows, and never use one for a variable that takes zero or negative values.

6.6 Colour, and the people who cannot see it

- About 8% of men and 0.5% of women have some form of colour vision deficiency, most often red and green. In a class of 120 engineering students that is roughly eight people.
- Your report will be printed in black and white, or photocopied. A figure that depends on hue dies there. Vary lightness and line style as well as hue, and check by printing in greyscale.
- Use hue for unordered categories, and a single hue running light to dark for ordered ones. Never use a rainbow scale for a quantity: it has no natural order, it creates false boundaries, and it is unreadable in greyscale.
- Three or four colours is usually the limit before a legend becomes a memory test. Label the lines directly instead of using a legend wherever you can.

6.7 Tables are a visualisation too

When the reader needs the exact values, when there are fewer than about twenty numbers, or when the numbers have different units, a well set table beats a chart. Right align numbers, use the same number of decimals down a column, order rows by something meaningful, and never report more digits than the measurement supports: a mean of 60 from seven students is 60, not 60.0000.

6.8 Dashboards

A dashboard is a graph that somebody has to read every day, which makes it harder, not easier.

- One question per panel, and at most five or six panels. A dashboard that shows everything is read by nobody.
- Every number needs a comparison beside it: last period, the target, or the other site. A naked number cannot be acted on.
- Date it. "Data to 24 October, updated daily at 06:00" prevents the most expensive mistake, which is acting on stale figures.
- Say what the viewer should do when a panel goes red. If nothing follows, the panel is decoration and should be removed.

6.9 A checklist before anybody else sees your graph

1. Can you say in one sentence what comparison this graph makes?
2. Are both axes labelled, with units?
3. Do bars start at zero? If an axis is cut, is that stated?
4. Is the sample size on the figure, or in the caption?
5. Have you plotted the data, rather than only a summary of it?
6. Is uncertainty shown, or honestly discussed in the caption?
7. Are the categories ordered by something that helps the reader?
8. Does it survive a greyscale photocopy?
9. Would it still be legible at the size it will actually be printed?
10. Could a reader who disagrees with you check your claim from this picture?

6.10 Doing it on a computer

In R, `ggplot2` builds a graph by mapping variables to channels, which is exactly the vocabulary of this chapter: `aes(x =, y =, colour =)` is a set of channel choices. In Python, `matplotlib` gives

you full control and `seaborn` gives you sensible defaults for distributions and groups. Spreadsheet charts are acceptable for exploration and are hard to make honest for publication: they default to the chart types this chapter warns about.

Whatever you use, write the figure as code in a script, not by clicking. A figure you cannot rebuild from a script is a figure you cannot correct when the data change, and the data always change.

6.11 Glossary

Visual channel	the perceptual property that carries the number: position, length, angle, area, shade or hue
Small multiples	the same chart repeated for each group, on identical axes
Truncated axis	an axis that does not start at zero, which makes bar lengths meaningless
Dual axis	two vertical scales on one chart, which can manufacture an apparent relationship
Log scale	an axis on which equal distances represent equal ratios
Sequential palette	one hue from light to dark, for an ordered quantity
Data-ink	the part of the ink on a chart that carries information, as opposed to decoration

6.12 Summary

Week 6 in six lines

1. Decide the comparison before you choose the chart.
2. The eye reads position best, then length, then angle, then area, then colour.
3. Plot the data, not just a bar of its mean.
4. Bars start at zero; lines and dots need only an honest, labelled scale.
5. Every common chart trick pushes towards a stronger claim, never a weaker one.
6. A figure that does not survive greyscale, or that nobody can check, is not finished.

Exercises

1. A manager shows a bar chart of monthly plant availability with the vertical axis running from 96% to 98%, and calls it a dramatic improvement. The four values are 96.2, 96.8, 97.1 and 97.4. State what is wrong, redraw it in words, and say what you would put in the caption if you kept the cut axis.

2. Two shifts have mean output 412 and 418 tonnes, from 9 days each. Your colleague draws two bars. Describe a better display, and say what the reader can learn from yours that they cannot learn from his.

3. A bubble chart shows exports of two minerals, with radii of 1 cm and 2 cm. The caption says the second is twice the first. What does the picture actually say, and what radius would have been correct?

4. A chart of ore grade over 20 years uses a left axis for grade and a right axis for the gold price, and the two curves track each other closely. Give two reasons this is weak evidence of a relationship, and propose a better display.

5. Cases of a disease are 1.2, 2.4, 4.8, 9.6 and 19.2 thousand in five successive years. Sketch the linear and the logarithmic versions, and say which you would put in front of a minister, and why.

6. You have the exam marks of 240 students in six programmes and you want to compare the programmes. Propose a display, defend it against two alternatives, and state what you would do differently if there were 24 programmes.

7. Take any chart from a newspaper, a company report or a government dashboard. Apply the ten point checklist and report which points it fails. Redraw it.

8. A dashboard panel shows "Availability 97.4%" in large green type and nothing else. List three things that must be added before the number can be acted on.

Answers to the odd-numbered exercises

1. The bars encode length, so cutting the axis at 96 makes the last bar about four times the first for a difference of 1.2 percentage points. Redraw as bars from zero, or better, as a line or dot plot over the four months with the axis labelled 96 to 98 and the target marked, since the question is about a small change. Kept as a cut axis, the caption must say: "note the vertical axis begins at 96%; the total change is 1.2 percentage points."

3. Area grows as the square of the radius, so doubling the radius multiplies the area by four: the picture says four times, not twice. For a doubling, the radius should be multiplied by $\sqrt{2} \approx 1.41$, that is 1.41 cm. Better still, use a bar chart, where length is read accurately.

5. On the linear axis the first three years look flat and the last year dominates; on the log axis the five points are a straight line, because each year is exactly double the last and $\log_{10} 2 = 0.301$ each step. Give the minister both: the linear one to show the scale of what is coming, the log one to show that the growth rate has not changed. Label the log axis in plain numbers, 1, 2, 4, 8, 16 thousand.

7. Open ended. A good answer names the specific failures, for example no sample size, no units, alphabetical ordering, a cut axis without a note, and produces a redraw that fixes them without inventing data the original did not have.

Before next week

Find one graph you have made yourself, in any course, and run the ten point checklist over it honestly. Bring the original and your redraw. Next week we move from describing data to reasoning about chance, and the pictures become probability distributions.